Görkem Tekin

Master's Student

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PROFILE

I'm a master's student in Computational Modeling and Simulation at TU Dresden, specializing in Visual Computing. My work sits where computer vision, graphics and simulation meet: reconstructing objects from touch and vision, generating synthetic data for perception, and building virtual environments for testing.

EDUCATION

M.sc Computational Modeling and Simulation 09/2025 – Present
TU Dresden Dresden

BS Computer Education and Educational Technology 09/2020 – 06/2025
Bogazici University Istanbul

PROFESSIONAL EXPERIENCE

Map Creation and Simulation Engineering 09/2026 – Present
Fraunhofer IVI Dresden

- Build and maintain photorealistic HD map environments for the **CARLA** autonomous-driving simulator, covering the full pipeline from surveyed road data to a drivable simulation: **Trian3DBuilder** export → **Blender** / **Datasmith**/ FBX import into **Unreal Engine 4** → **ASAM OpenDRIVE** logical layer → validation and traffic simulation.
- Prepare 3D scene content for city-scale maps, building and vegetation meshes, traffic light placement and signal controllers.
- Integrate the OpenDRIVE road network with the 3D scene so that autopilot, spawn points and traffic become available.
- Built a **Python** toolchain for HD map validation and repair (OpenDRIVE/XML parsing, lane-graph and junction checks, automated patching with before/after reporting), now the standard import step for new maps.

Autonomous Driving Software Engineer 11/2025 – Present
ELBFLORACE Formula Student Team TU Dresden e.V. Dresden

- **Architecting an R&D pipeline** using **BlenderProc** to automate the generation of high-fidelity synthetic datasets, significantly accelerating **YOLO** model training for track and cone detection.
- **Developing Digital Twins** of racing tracks and obstacles in **Blender**, utilizing **Python** and **C++** to simulate diverse environmental conditions for robust perception testing.
- **Contributing to the technical advancement** of a driverless racing vehicle currently **ranked 5th globally** in the Formula Student e.V world rankings.

Python and Robotics Instructor 10/2023 – 06/2024
Istanbul Metropolitan Municipality

- Instructed middle and high school students in **Python and C programming** fundamentals, facilitating practical coding exercises and project-based learning.
- Mentored and oversaw **robotics projects**, introducing students to technical innovation and **hardware-software integration** through hands-on guidance.

LANGUAGES

English
C1

German
A2